Saksham Jain

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Education

PhD, Statistics Sep 2022 – Present
University of Washington, Seattle
Advisors: Alex Luedtke, Ting Ye

MS, Electrical and Computer Engineering Aug 2020 – May 2022
Duke University
Advisors: Cynthia Rudin, Alex Volfovsky, Sudeepa Roy

BE, Manufacturing Processes and Automation Engineering Aug 2015 – Jun 2019
Netaji Subhas Institute of Technology, University of Delhi
Advisor: Umang Soni

Research Interests

Causal inference; semiparametric efficiency theory; deep learning; interactive machine learning; adaptive experimentation; machine learning for scientific applications, particularly in biology and public health

Peer-Reviewed Publications

* denotes equal contribution.

1. **Saksham Jain** and A. Luedtke. Conditional Distributional Treatment Effects: Doubly Robust Estimation and Inference. In *Forty-third International Conference on Machine Learning (ICML)*, 2026.
2. **Saksham Jain***, T. M. Seale-Carlisle*, C. Lee, C. Levenson, S. Ramprasad, B. Garrett, S. Roy, C. Rudin, and A. Volfovsky. Evaluating pre-trial programs using machine learning matching algorithms. In *AAAI*, 2024 (oral).
3. C. Kornfein, **Saksham Jain***, F. Willard*, C. Tang*, Y. Long*, J. Malof*, S. Ren*, and K. Bradbury*. Closing the Domain Gap: Blended Synthetic Imagery for Climate Object Detection. In *NeurIPS, Tackling Climate Change with Machine Learning Workshop*, 2022. Extended version: *Environmental Data Science*, 2023.
4. G. K. Nayak, K. R. Mopuri, **Saksham Jain**, and A. Chakraborty. Mining Data Impressions from Deep Models as Substitute for Unavailable Training Data. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2021.
5. **Saksham Jain**, G. Seth, A. Paruthi, U. Soni, and G. Kumar. Synthetic data augmentation for surface defect detection and classification using deep learning. *Journal of Intelligent Manufacturing*, 1–14, 2020.
6. G. K. Nayak, **Saksham Jain**, R. V. Babu, and A. Chakraborty. Fusion of Deep and Non-Deep Methods for Fast Super-Resolution of Satellite Images. In *IEEE International Conference on Multimedia Big Data (BigMM)*, pages 267–271, 2020.
7. S. Jain, **Saksham Jain**, and A. P. Komanduri. A cascade predictive control strategy for active suspension systems. In *IEEE International Conference on Automation/XXIII Congress of the Chilean Association of Automatic Control (ICA-ACCA)*, pages 1–5, 2018.

Preprints and Working Papers

1. **Saksham Jain** and T. Ye. Longitudinal causal inference with multiple continuous exposures: an application to proteomics. *Working Paper*, 2026.

Research Positions

Health Care Policy, Harvard University, Research Intern, PI: Alex Luedtke, *MA, USA* — Jun–Sep 2026

RIKEN-AIP, Research Intern, PI: Masaaki Imaizumi, *Tokyo, Japan* — Jun–Sep 2025

Duke Energy Initiative (Bass Connections), Graduate Researcher, *NC, USA* — Aug 2021–May 2022

Almost Matching Exactly Lab, Duke University, Research Assistant, *NC, USA* — May 2021–Apr 2022

RAILabs, Duke University, Research Intern, PI: Joseph Lo, *NC, USA* — May–Aug 2021

Visual Computing Lab, IISc Bangalore, Research Assistant, *Karnataka, India* — Jan–Jul 2020

Selected Talks and Presentations

• Invited talk ⋆ Oral presentation ◇ Poster

Conditional Distributional Treatment Effects: Doubly Robust Estimation and Testing

- • CFE-CMStatistics, HTW Berlin, University of Applied Sciences, *Berlin, Germany* — Dec 2026
- • Department of Health Care Policy, Harvard Medical School, *Boston, MA, USA* — Oct 2026
- • Seminar, RIKEN Center for Advanced Intelligence Project, *Tokyo, Japan* — Jul 2026
- ◇ International Conference on Machine Learning (ICML), *Seoul, South Korea* — Jul 2026
- ⋆ Western North American Region of the IBS (WNAR) Meeting, *Pullman, WA, USA* — Jun 2026
- ◇ American Conference on Causal Inference (ACIC), *Salt Lake City, UT, USA* — May 2026

Evaluating pre-trial programs using machine learning matching algorithms

- ◇ Visit Day, Department of Statistics, University of Washington, *Seattle, WA, USA* — March 2024
- ⋆ Intern Presentations, Amazon Supply Chain Optimization Technologies, *New York City, USA* — Jul 2024
- ◇ Department of Biostatistics, University of Washington, *Seattle, WA, USA* — Feb 2024

Honors and Awards

Runner up, Outstanding Written Paper Award, Meeting of WNAR of the IBS — Jun 2026

Student Research Award, New England Statistics Symposium (NESS) 2026 — May 2026

Graduate Student Conference Presentation Award, University of Washington — Apr 2026

Coursera Department Fellowship, Department of Statistics, University of Washington — Sep 2022

CVSPK Talent Incentive Scheme, Netaji Subhas Institute of Technology, University of Delhi — Aug 2021
Awarded for outstanding undergraduate research

Merit Award, Computer Vision Summer School, IIIT Hyderabad — Jul 2019

Merit Award, Machine Learning Summer School, IIIT Hyderabad — Jul 2019

Bachelor's Thesis Ranked 2, MPAE Division, Netaji Subhas Institute of Technology, Univ. of Delhi — May 2019

Finalist, American Astronautical Society CanSat Competition 2018, TX, USA — Jun 2018

Teaching Experience

University of Washington

STAT 509 – Econometrics I: Introduction to Mathematical Statistics	Autumn 2025
STAT 451 – Visualizing Data	Autumn 2024
STAT 435 – Introduction to Statistical Machine Learning	Spring 2023
STAT 390 – Statistical Methods in Engineering and Science	Winter 2023
STAT 311 – Elements of Statistical Methods	Fall 2022

Duke University

STA 561D – Probabilistic Machine Learning (Statistics)	Spring 2022
ECE 586 – Vector Space Methods with Applications (ECE)	Fall 2021

Industry Experience

Amazon Science, Applied Scientist II Intern, *NYC, USA* Jun–Sep 2024

- Fine-tuned LLMs within deep sequence models for multi-horizon time-series forecasting.
- Integrated text embeddings from pre-trained sentence transformers to improve forecasting performance for diverse product categories.

involve.ai, AI/ML Engineering Intern, *LA, USA* Jul–Sep 2023

- Built a research-and-revise approach for post-hoc LLM output correction, implemented as a co-pilot for market research tasks.
- Built a DistilBERT multi-task classifier for job titles, improving recall for sales lead retrieval by 4×.

Million Sparks Foundation, Machine Learning Engineer, *UP, India* Jun 2019–Jan 2020

- Developed an unsupervised domain adaptation network for real-time user outcome prediction, achieving $\sim 4.3\%$ improvement over state-of-the-art AUC scores.
- Built models for text classification using BERT, recommendation using neural collaborative filtering, and explainable churn prediction with SHAP.

Professional Service

Reviewer, Statistics in Medicine, 2026

Reviewer, Annals of Applied Statistics, 2026

Service and Leadership

Pre-application Review Service (PARS), Volunteer, UW Seattle.	Sep–Dec 2022
ECE Representative, Graduate and Professional Student Council, Duke University.	Aug 2020–Jan 2021
Math and Astronomy Society, Lead, Mathematics, NSIT, University of Delhi.	Aug 2017–Sep 2018
Dramatics Society, Lead, Scriptwriting, NSIT, University of Delhi.	Aug 2016–May 2018